

Zomato

Increasing no. of text reviews for the food delivery vertical

Team: Ratings and Reviews

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Status: In Development

PROBLEM DEFINITION

Users rely on reviews before ordering food, but writing them is uncommon. This is a problem because the online food delivery market is booming, with revenue expected to reach billions by 2025. Restaurants with strong reviews will attract more customers, leading to higher revenue.

- **What is the problem?**

Customers do not write reviews because they find the process complicated, time consuming and boring. As a result, there is a reduction in no. of text reviews for users seeking comprehensive feedback for restaurants.

- **Who is facing the problem?**

The customers who order at least four times a month seeking reliable reviews. These groups depend on comprehensive reviews to make informed decisions and enhance their experience.

- **What is the business value that will be unlocked by solving the problem?**

More number of quality reviews increases conversion rate, thereby increasing the revenue.

Reviews helps users in making informed decisions, hence decreasing order mistakes which leads to reduction in refunds and returns.

Reviews help in enhancing user experience, therefore increasing the customer retention rate,

- **How will the target users benefit if the problem is solved?**

With more reviews, users can make informed choices for their meal, selecting restaurants and dishes based on food quality, service, and value, ultimately avoiding disappointing experiences.

Users uncover hidden restaurant favourites based on real user experiences with the help of reviews.

- **Why is it urgent to solve this problem now?**

Solving this problem now is crucial as Zomato turned profitable in 2023. With food delivery revenue growing by 29% in Q3 FY24 to INR 2,205 Cr, more reviews from frequent users can build on this success, attracting more customers and boosting trust.

GOALS

- *To increase the percentage of orders reviewed per reviewer per month*
- *To increase the no. of reviewers*
- *To decrease the drop off rate in review writing process*

Functional metrics-

1. Average number of reviews per reviewer per month
2. Percentage of orders that are reviewed per reviewer
3. Average number of orders per reviewer per month
4. Percentage of users leaving reviews
5. No. of new reviewers onboarded per month
6. No. of reviewers who continue writing reviews over time (retention rate)
7. Average time taken to complete a review
8. Percentage of reviewers who start but do not complete the review process

Non- Functional Metrics-

1. System's ability to handle the increase in no. of reviews and reviewers
2. Number of system errors that occurred during the review process.
3. Average time taken by the system to respond to user actions

● **Why are these metrics important?**

These metrics are important as they collectively gauge the performance, efficiency, scalability, and success of systems and the review process. They also play a crucial role in measuring user engagement with the platform and provide valuable feedback loops for continuous improvement.

NON-GOALS

- Marketing efforts for the review process are not included.
- Review quality is not a focus at this point of time.
- Adding a few features to the review process is within scope, but not the complete redesigning of the process.
- Providing incentives or discounts is not included.

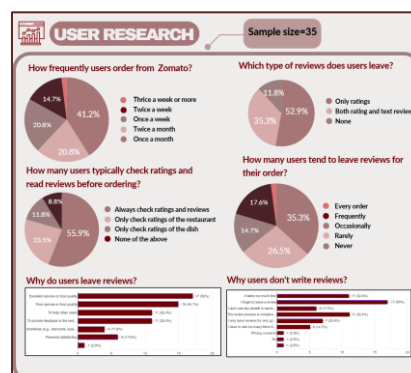
VALIDATION OF THE PROBLEM

A survey with survey size of 35 people and 4 1:1 user interviews reveal that while users prefer reading reviews before placing orders, they lack the motivation to write reviews themselves.

According to Zomato's Blog, only 1% of users write reviews.

● **Insights from user research/data**

According to survey data, around 35% of users have never written a review and ~26% of users rarely do so. Therefore, we can say that this is a real problem!



UNDERSTANDING THE TARGET AUDIENCE

• User Segment

Who? Users who order at least 4 times a month

Why?

Higher impact: This segment's frequent ordering positions them as a powerful group for increasing review numbers; even if they review fewer orders, their high frequency will still significantly boost overall reviews.

Familiarity with the app: They are likely familiar with the app and its functionalities, making the review process easier for them.

Better insights: Their varied experiences with different orders provide better insights into the overall quality and performance of the restaurants and platform.

Size of the segment?

Zomato's MAUs= 18M

Around 40% of users order a minimum of 4 times a month

No. of targeted users= 7.2 M

• Key Personas



Shweta Verma, 25, Female

Location- Gurugram

Occupation- Marketing Executive

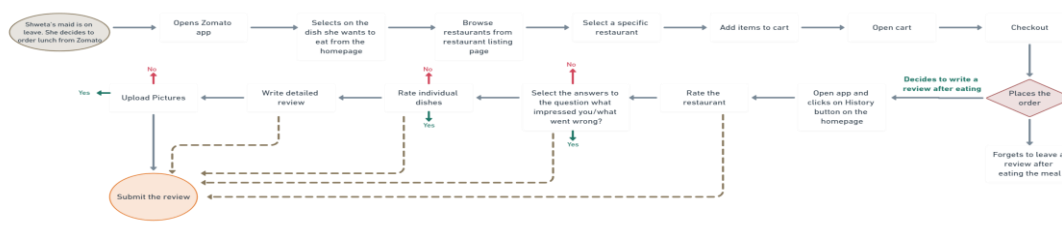
Income- 15 LPA

"I order food from Zomato whenever my cook doesn't come or when I am craving for a specific food item"

She is a frequent user, but rarely leave a text review for her order.

Pain Points	Needs
• She is not motivated to write reviews and finds the process boring and complicated. • She forgets to leave a review after eating the meal. • Uncertain about what constitutes a helpful or impactful review.	• She needs to know how her reviews are helping others. • She wants reminders from the app to write a review. • She needs tips and templates to help structure reviews, making it easier to know what to write.

• User Flow



This user journey flow depicts Shweta's process of ordering food on Zomato and the steps involved in leaving a review.

SOLUTION

We came up with the solution of AI-powered prompt reviews after hypothesizing that personalized and dynamic prompts would significantly increase review rates.

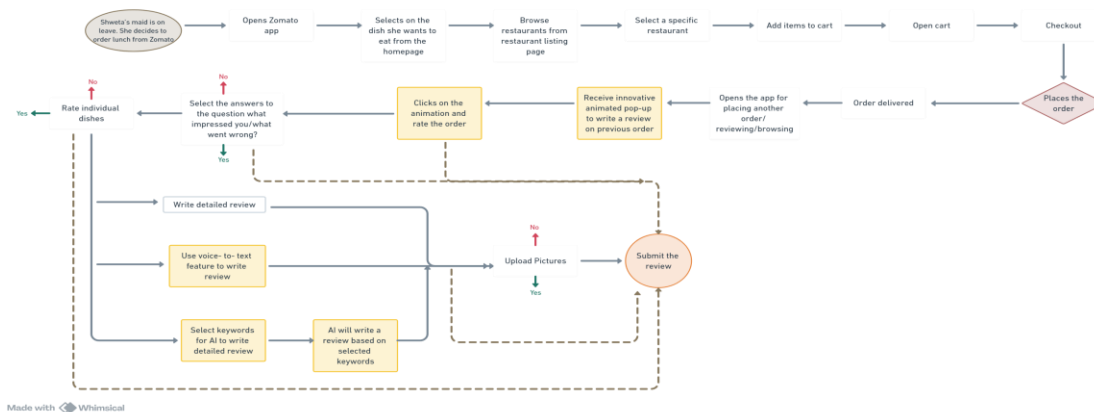
See appendix for the whole solution hypothesis process.



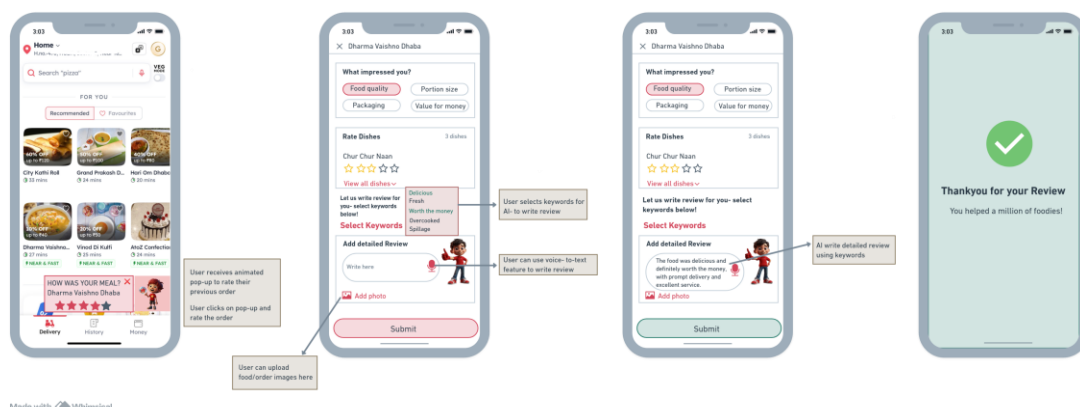
What is this solution? Users will receive an engaging, animated pop-up on the homepage, prompting them to review their recent order, only if the previous order was placed within a week. The AI will leverage keywords to generate detailed and personalized review suggestions. Additionally, a voice-to-text feature will be available, allowing users to verbally provide their reviews, ensuring a seamless and user-friendly experience.

Why this solution? User surveys and interviews revealed that users often forget to leave reviews and find the process boring and complicated. This solution addresses these issues by displaying an engaging, animated pop-up on the homepage, reminding users to leave a review. The AI-generated review suggestions simplify the process, and the voice-to-text feature makes it easy for users to provide their feedback verbally.

User Flow with Proposed Solution



Wireframes



- **Key Features**

1. **In-app Pop-Up-** Animated prompts within the app encouraging users to rate their previous order upon opening.
2. **Personalized AI- Powered Review Prompts-** Prompts are generated based on user history, order context and selected keywords.
3. **Unique Review Generation-** Algorithms that prioritize varying sentence structures, perspectives, and focal points to be implemented so that AI produces unique and diverse reviews despite using similar set of keywords.
4. **Voice-to-text Feature** - Allows users to submit reviews using voice input.

- **Schema Changes**

1. **User Table-** To store user data. No changes to this table

Users

Column Name	Data Type	Description
user_id	INT	PK, unique identifier
name	VARCHAR(50)	User's name
email	VARCHAR(100)	User's email, unique

2. **Order Table-** Add new columns to track the status of review for each order.

Orders

Column Name	Data Type	Description
order_id	INT	PK, unique identifier
user_id	INT	FK Users (user_id)
order_date	TIMESTAMP	Date and Time of Order
review_status	BOOLEAN	Order reviewed- True or False

3. **Popup Table-** Logs each instance of a popup shown to a user. Star ratings are stored in the existing schema.

Pop-ups

Column Name	Data Type	Description
user_id	INT	FK Users (user_id)
order_id	INT	FK Orders (order_id)
popup_sent	ENUM	Popup sent or not
popup_sent_at	TIMESTAMP	Date and time popup shown
popup_interaction	ENUM	'rated', 'dismissed', 'ignored'

4. **Review Keywords Table-** Stores pre- defined keywords for reviews.

Keywords

Column Name	Data Type	Description
keyword_id	INT	PK
keyword	VARCHAR(25)	Pre-defined keyword for reviews

5. **AI Reviews Suggestion Table-** Stores AI-generated review suggestions for each order.

AI Reviews

Column Name	Data Type	Description
review_id	INT	PK, unique identifier
user_id	INT	FK Users (user_id)
order_id	INT	FK Orders (order_id)
keyword_ids	ARRAY	FK Keywords (keyword_id)
suggested_review	TEXT	AI-generated review text

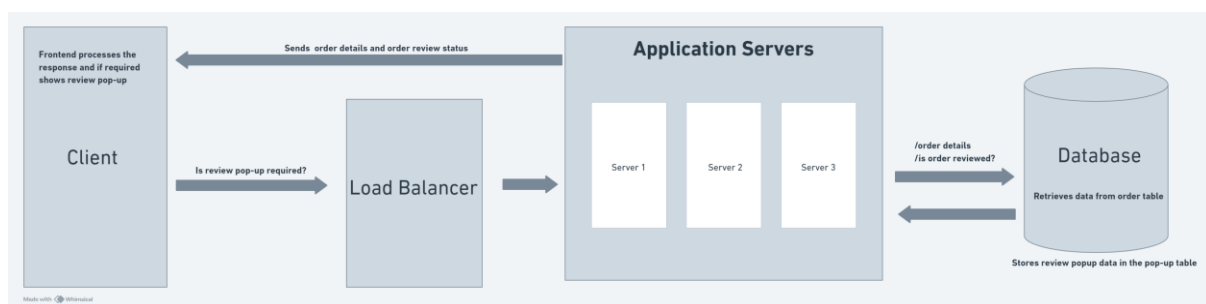
6. **Voice Reviews-** Voice-to-text reviews will be stored in the existing schema.

- **Algorithm Changes**

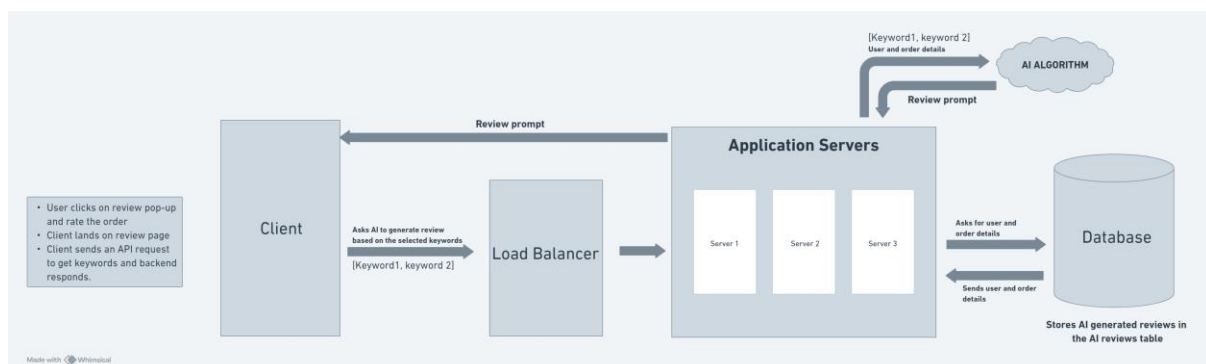
1. **N-gram Language Model-** Generate text by predicting the next word in a sequence, ensuring that generated sentences are natural and varied.
2. **Pre-trained Language Models-** Use advanced models like GPT-3 to generate diverse and contextually relevant sentences.
3. **Google Speech-to-Text API-** Integrate speech recognition to capture voice input.
4. **NLP Models-** Use NLP to process and convert voice input into text.

- **System Design**

Receiving Review Pop-up



Generating AI Review



- **Key metrics**

North Star Metric-

- No. of reviews written using keywords and AI / total no. of reviews
- No. of reviews written using voice assistant/ total no. of reviews

L1 metric-

- No. of users interacting with the pop-up / total no. of users
- Average time taken to complete the review process with the proposed solution
- No. of users writing reviews using AI or voice assistant / no. of user interacting with the pop-up

- Increase in avg. no. of reviews per user per month after implementation of new features
- Diversity index of reviews
- NPS score
- **Edge/ Error Case Handling**
 - **AI Failure to write a review-**
Display a generic prompt- "Oops! We couldn't generate your review this time. Please share your experience with us manually. Your feedback helps us improve!"
 - **Inability to show or select keywords**
Users can write review manually
 - **Microphone Access Denied**
Prompt users to enable microphone access when they click on microphone.
 - **Voice Recognition System Failure**
Users can write review manually
- **Marketing**
 1. **Social Media Announcements-** Use Zomato's social media pages to announce the new feature.
 2. **In-app Short Videos-** Short videos within the app guiding users on how to use the new feature.

LAUNCH READINESS

- **Steps Leading Up to the Launch**
 1. **Requirement Gathering and Planning:**
 - Define project scope and objectives.
 - Conduct user research and gather requirements.
 - Identify key stakeholders and form project teams.
 2. **Design Phase:**
 - Create wireframes and prototypes.
 - Develop user flow diagrams.
 - Review and approve final design.
 3. **Development Phase:**
 - Backend development: Implement AI algorithms, voice recognition, and database setup.
 - Frontend development: Integrate UI/UX design, animated pop-ups, and voice-to-text feature.
 - Integration and testing: Combine backend and frontend, conduct initial functionality tests.
 4. **Quality Assurance (QA) Phase:**

- Conduct thorough QA testing to identify bugs and usability issues.
- Perform User Acceptance Testing (UAT) with a select group of users.
- Make necessary refinements based on feedback.

5. Dogfooding Phase:

- Roll out the solution to internal teams for real-world usage.
- Gather feedback from internal users and address any issues.

6. Launch Phase:

- Soft launch to a small segment of users to monitor performance (A/B testing).
- Full launch to all users after successful soft launch.

• Key Milestones

Milestones	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Design Complete															
Development Complete															
QA Complete															
Dogfooding Complete															

*W= Week

• Internal Stakeholders

- Product Management Team
- Engineering Teams
- QA Teams
- Customer Support
- Marketing and Communications
- Data Analytics

RISKS AND MITIGATION

1. **User Annoyance from Frequent Prompts**

Mitigation: Show pop-ups only once for the most recent order.

2. **Inaccurate or Irrelevant AI Suggestions**

Mitigation: Regularly update the AI with new data and use user feedback to improve its suggestions.

3. **Resistance to New Features**

Mitigation: Offer simple tutorials and highlight the benefits to make users comfortable with the new features.

4. **Misinterpretation of Voice-to-Text Inputs**

Mitigation: Enhance the accuracy of the voice-to-text engine by continuously training it with diverse voice samples.

5. **Scalability Risk with Large Number of Prompt Requests, Voice Recognition, and Database Interactions**

Mitigation: Implement cloud-based scalable architecture, caching, and load testing to handle increased user engagement.

FUTURE ITERATIONS

- Scope for Next Versions

1. AI-Generated Keywords: Train the AI to automatically generate relevant keywords based on the user's previous orders and feedback trends.

Benefit: Enhances the personalization and relevance of review prompts, leading to more meaningful user feedback.

2. Multilingual Support: Expand Google speech-to-text API and AI review prompts to support local and cultural languages.

Benefit: Broadens the user base by catering to non-English speakers, improving accessibility and user satisfaction.

3. Advanced Sentiment Analysis: Implement sentiment analysis to categorize reviews as positive, negative, or neutral, and adjust future prompts accordingly.

Benefit: Provides deeper insights into user feedback and improves the quality of AI-generated review suggestions.

4. Dynamic and Seasonal/Trend-Based Prompts in the Review Pop-up: By rotating through different types of prompts (e.g., taste, service, overall satisfaction) and integrating seasonal or trend-based prompts (e.g., special menu items during festive seasons), users receive varied and relevant prompts based on their recent experiences and current events.

Benefit: This approach enhances user engagement by providing fresh and personalized prompts, encouraging more frequent and detailed feedback.

- Ideas Parked for Later

1. Community Q&A Forum: Introduce a forum where users can ask and answer questions about restaurants, fostering engagement and indirectly encouraging reviews.

2. Redeemable Coupons and Verified Badges: Users receive redeemable coupons from partner/collaborator brands for writing reviews. They also earn verified badges after writing a certain number of reviews, incentivizing consistent participation.

3. Review Summarization: Automatically generate summaries of reviews for each restaurant, highlighting common themes and sentiments, providing users with quick insights into the overall feedback for a restaurant and prompt them to leave their own reviews based on summarized insights.

4. Interactive Polls: Gather quick feedback through polls on specific aspects of dining experiences, then encourage users to provide detailed reviews to enrich insights.

APPENDIX

1. Read more about the competitive analysis - [Market research](#)
2. Read more about user research and data analysis- [User research](#)
3. Read more about the alternate solutions considered- [Solution](#)